



CS 431/631 Introduction to Big Data

Basic Data Processing and Analysis

Some slides are borrowed from Vipin Kumar, et. al.

Lei Yang
Computer Science & Engineering
University of Nevada, Reno



Summary Statistics

- Summary statistics are numbers that summarize properties of the data
- Summarized properties include frequency, location and spread
 - Examples: location – mean, median
 spread - standard deviation, variance

Summary Statistics

- Consider a sample $x_1, x_2, \dots, x_n$

- Sample mean $\bar{x} = \frac{\sum_{j=1}^n x_j}{n},$

- Sample variance $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1},$

```
def two_pass_variance(data):
```

```
    n = len(data)
```

```
    mean = sum(data) / n
```

```
    variance = sum([(x - mean) ** 2 for x in data]) / (n - 1)
```

```
    return variance
```

Summary Statistics

*Most summary statistics can be calculated in a **single pass** through the data*

Welford's One-pass Algorithm

$$\bar{x}_n = \frac{(n-1)\bar{x}_{n-1} + x_n}{n} = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n}$$

$$s_n^2 = \frac{n-2}{n-1} s_{n-1}^2 + \frac{(x_n - \bar{x}_{n-1})^2}{n} = s_{n-1}^2 + \frac{(x_n - \bar{x}_{n-1})^2}{n} - \frac{s_{n-1}^2}{n-1}, \quad n > 1$$

Frequency and Mode

- The *frequency* of an attribute value is the **percentage of time** the value occurs in the data set
 - For example, given the attribute 'gender' and a representative population of people, the gender 'female' occurs about 50% of the time.
- The *mode* of an attribute is the **most frequent attribute** value
- The notions of frequency and mode are typically used with **categorical data**

Frequency and Mode

- **Example 1: Find the mode of the given data set:**
3, 3, 6, 9, 15, 15, 15, 27, 27, 37, 48.
 - **Solution:** 15 is the mode since it is appearing more number of times in the set compared to other numbers.
- **Example 2: Find the mode of 4, 4, 4, 9, 15, 15, 15, 27, 37, 48 data set.**
 - **Solution:** both the number 4 and 15 are modes of the set.
- **Example 3: Find the mode of 3, 6, 9, 16, 27, 37, 48.**
 - **Solution:** If no value or number in a data set appears more than once, then the set has no mode.

Percentiles

- For continuous data, the notion of a percentile is more useful.

Given an ordinal or continuous attribute x and a number p between 0 and 100, the p th percentile is a value x_p of x such that $p\%$ of the observed values of x are **less than x_p** .

- For instance, the 50th percentile is the value $x_{50\%}$ such that 50% of all values of x are less than $x_{50\%}$.

Percentiles

- Example: You are the fourth tallest person in a group of 20
 - 80% of people are shorter than you:



- That means you are at the 80th percentile.
- If your height is 1.85m then "1.85m" is the 80th percentile height in that group, i.e., $x_{80\%} = 1.85\text{m}$

Measures of Location: Mean and Median

- The mean is the most common measure of the location of a set of points.
- However, the mean is very sensitive to outliers.
- Thus, the median or a trimmed mean is also commonly used.

$$\text{mean}(x) = \bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$$

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r + 1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

For example, if we have set of values = 2,2,3,4,5, then;

Mean = (2+2+3+4+5)/5 = 3.2

Median = 3

Mode = 2

Measures of Spread: Range and Variance

- Range is the difference between the max and min
- The variance or standard deviation s_x is the most common measure of the spread of a set of points.

$$\text{variance}(x) = s_x^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2$$

- Because of outliers, other measures are often used.

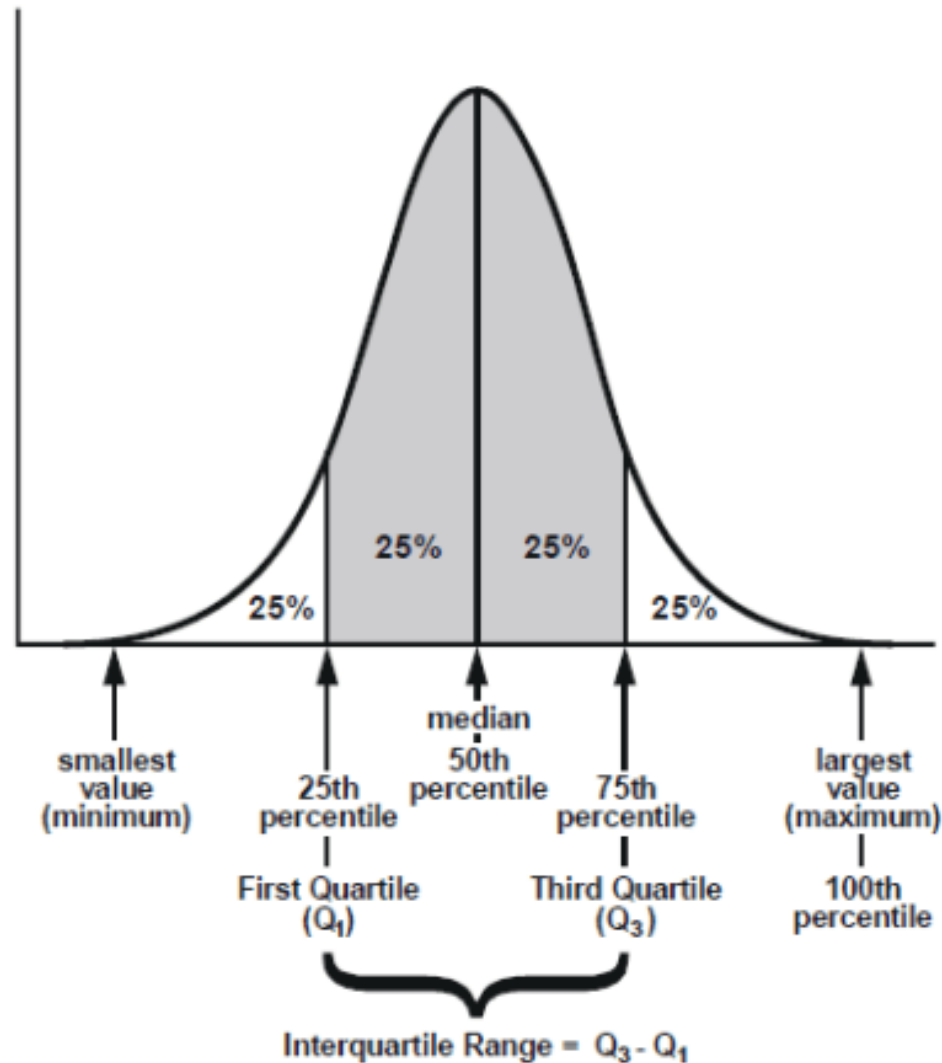
$$\text{AAD}(x) = \frac{1}{m} \sum_{i=1}^m |x_i - \bar{x}| \quad (\text{Absolute Average Deviation})$$

$$\text{MAD}(x) = \text{median}\left(\{|x_1 - \bar{x}|, \dots, |x_m - \bar{x}|\}\right) \quad (\text{Median Absolute Deviation})$$

$$\text{interquartile range}(x) = x_{75\%} - x_{25\%} \quad (\text{IQR, midspread, middle 50\%, H-spread})$$

Measures of Spread: Range and Variance

$$\text{interquartile range}(x) = x_{75\%} - x_{25\%}$$



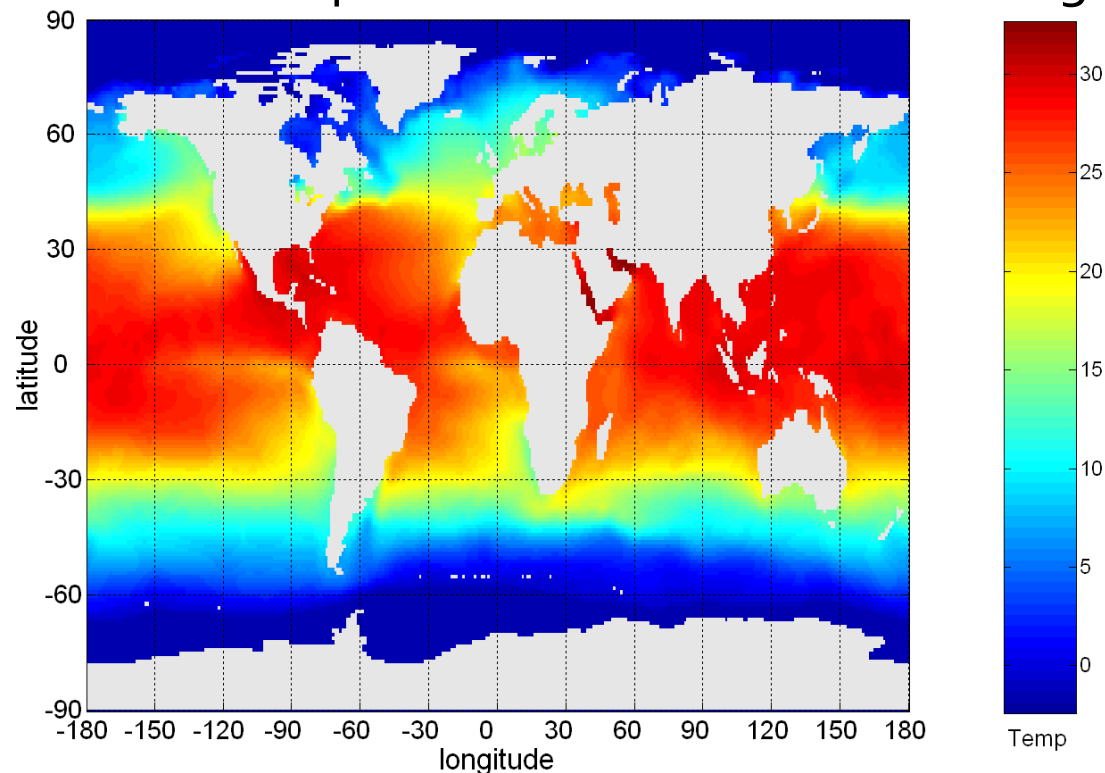
Visualization

Visualization is the conversion of data into a **visual or tabular format** so that the **characteristics** of the data and the **relationships** among data items or attributes can be analyzed or reported.

- Visualization of data is one of the most powerful and appealing techniques for data exploration.
 - Humans have a well developed ability to analyze large amounts of information that is presented visually
 - Can detect general **patterns and trends**
 - Can detect **outliers** and **unusual** patterns

Example: Sea Surface Temperature

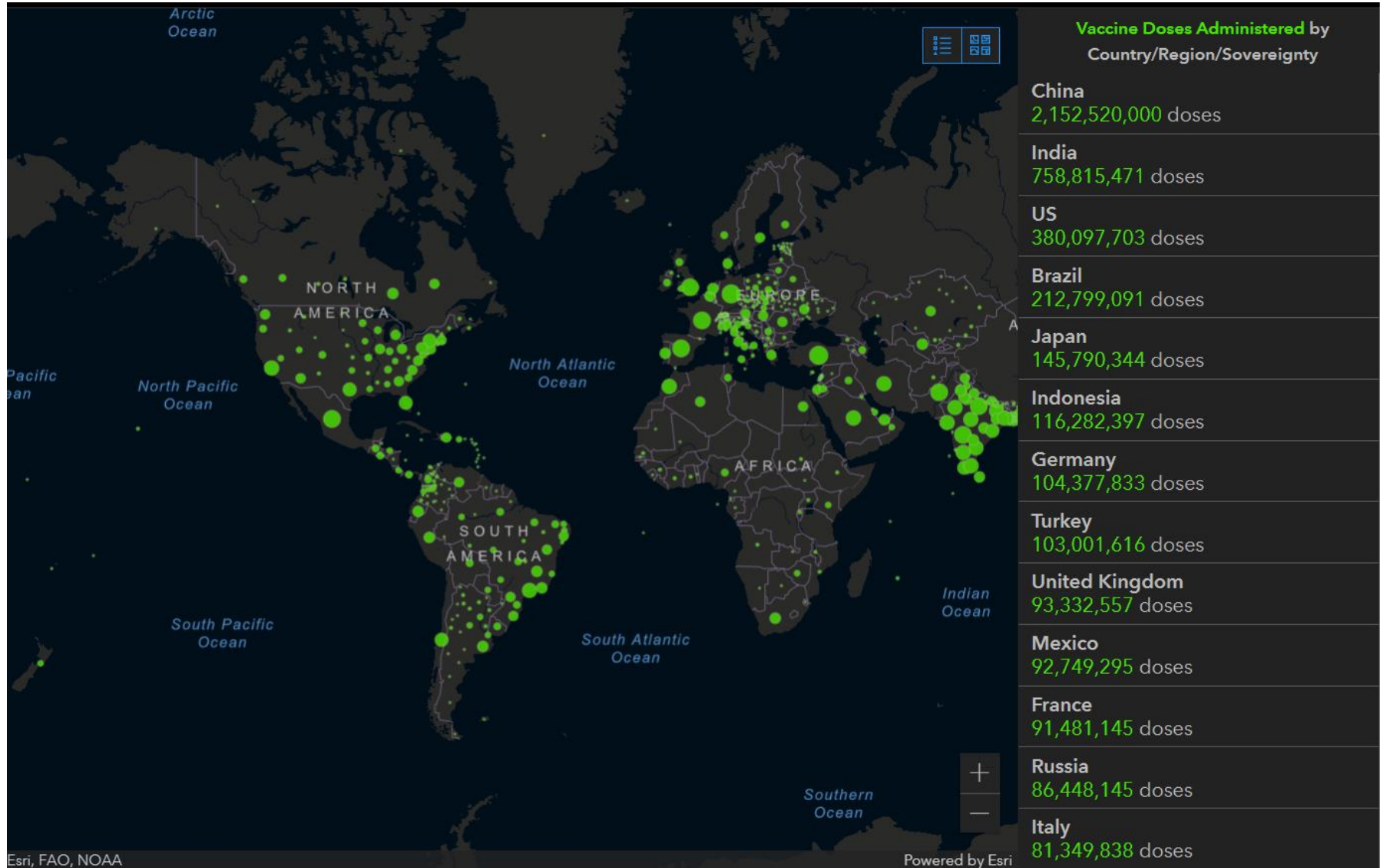
- The following shows the Sea Surface Temperature (SST) for July 1982
 - Thousands of data points are summarized in a single figure



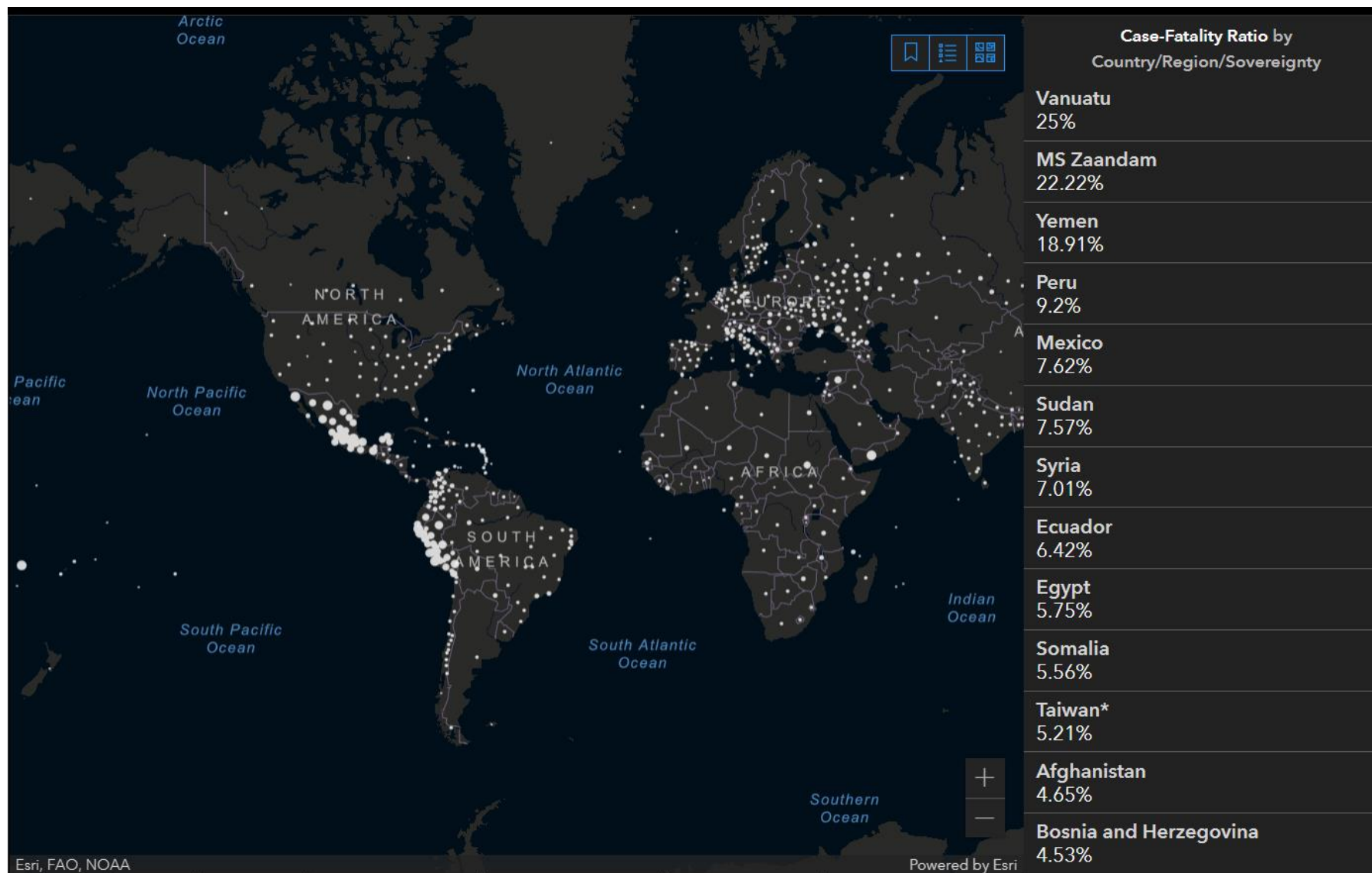
Cumulative cases as of 9/14/2021



Vaccinations as of 9/14/2021



Case-fatality ratio as 9/14/2021



Representation

- Is the mapping of information to a visual format
- Data objects, their attributes, and the relationships among data objects are translated into **graphical elements** such as points, lines, shapes, and colors.
- Example:
 - Objects are often represented as **points**
 - Their attribute values can be represented as the **position** of the points or the **characteristics** of the points, e.g., color, size, and shape
 - If **position** is used, then the relationships of points, i.e., whether they form groups or a point is an outlier, is easily perceived.

Arrangement

- Is the **placement** of visual elements within a display
- Can make a large difference in how easy it is to understand the data
- Example:

		<i>attributes</i>					
<i>objects</i>		1	2	3	4	5	6
	1	0	1	0	1	1	0
	2	1	0	1	0	0	1
	3	0	1	0	1	1	0
	4	1	0	1	0	0	1
	5	0	1	0	1	1	0
	6	1	0	1	0	0	1
	7	0	1	0	1	1	0
	8	1	0	1	0	0	1
	9	0	1	0	1	1	0

		<i>attributes</i>					
<i>objects</i>		6	1	3	2	5	4
	4	1	1	1	0	0	0
	2	1	1	1	0	0	0
	6	1	1	1	0	0	0
	8	1	1	1	0	0	0
	5	0	0	0	1	1	1
	3	0	0	0	1	1	1
	9	0	0	0	1	1	1
	1	0	0	0	1	1	1
	7	0	0	0	1	1	1

Selection

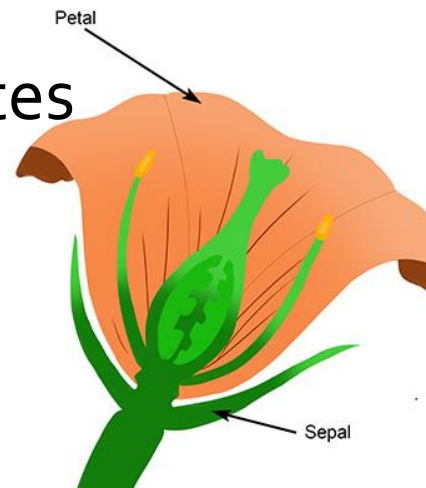
- Is the **elimination** or the **de-emphasis** of certain objects and attributes
- Selection may involve the choosing a subset of attributes
 - **Dimensionality reduction** is often used to reduce the number of dimensions to two or three
 - Alternatively, **pairs** of attributes can be considered
- Selection may also involve choosing a subset of objects
 - A region of the screen can only show so many points
 - Can sample, but want to preserve points in sparse areas

Iris Sample Data Set

- Many of the exploratory data techniques are illustrated with the Iris Plant data set.
 - Can be obtained from the UCI Machine Learning Repository <http://www.ics.uci.edu/~mlearn/MLRepository.html>
 - From the statistician Douglas Fisher
 - Three flower types (classes):
 - Setosa
 - Virginica
 - Versicolour
 - Four (non-class) attributes
 - Sepal width and length
 - Petal width and length



Virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland Science Institute.

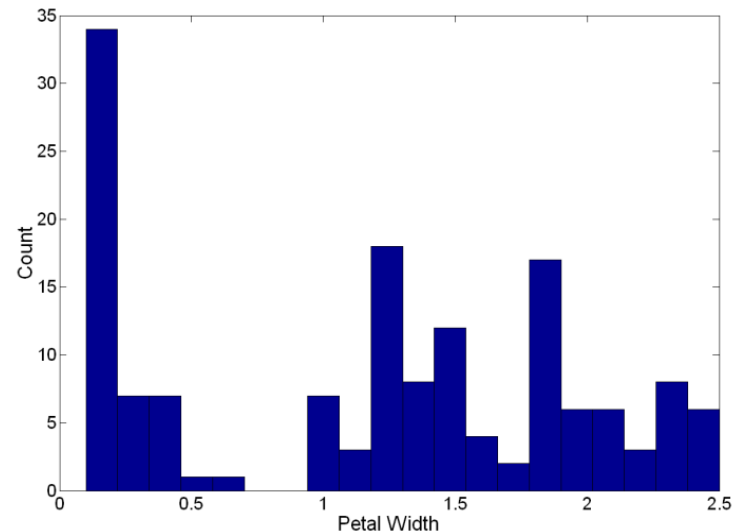
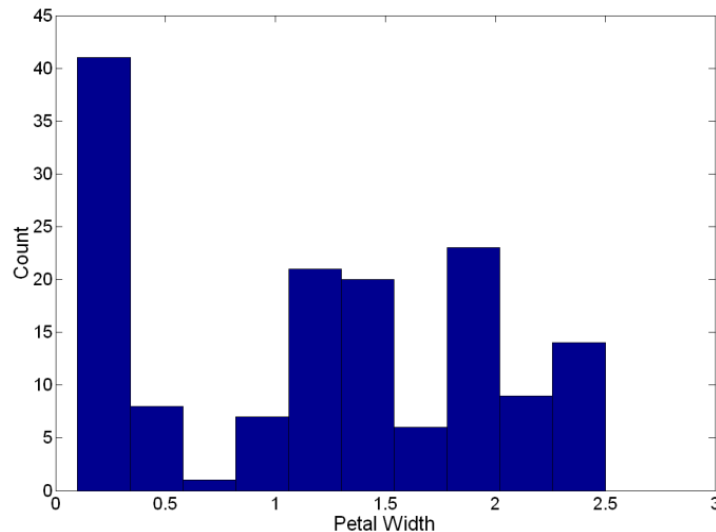


Visualization Techniques: Histograms

■ Histogram

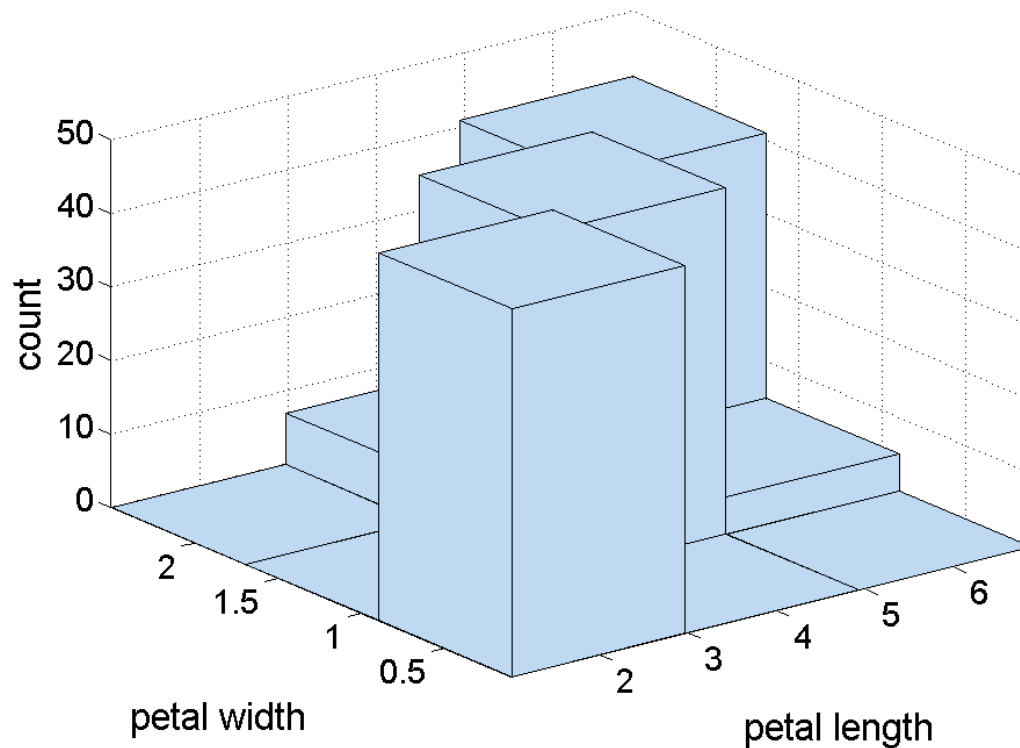
- Usually shows the **distribution** of values of a single variable
- Divide the values into **bins** and show a bar plot of the number of objects in each bin.
- The height of each bar indicates the number of objects
- Shape of histogram depends on the number of bins

■ Example: Petal Width (10 and 20 bins, respectively)



Two-Dimensional Histograms

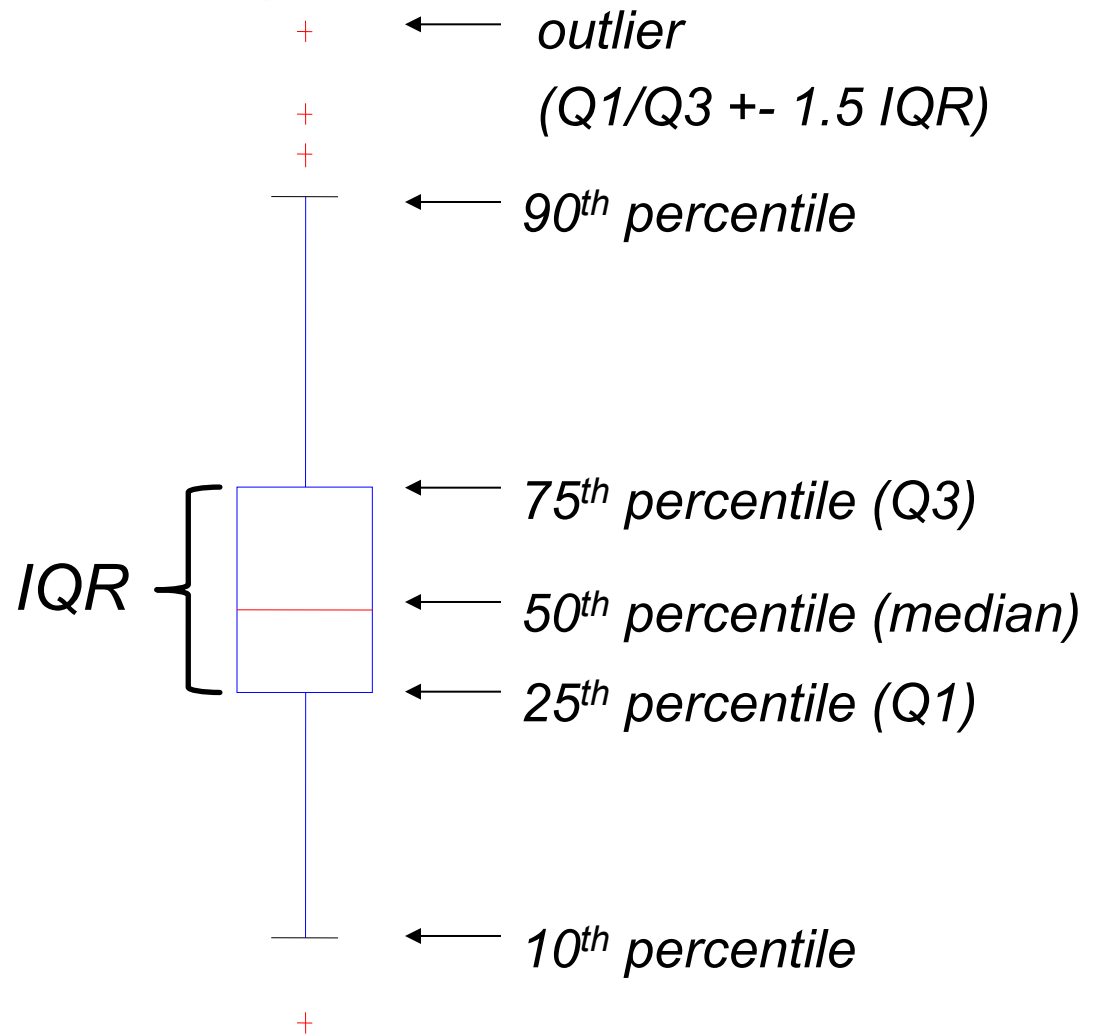
- Show the joint distribution of the values of two attributes
- Example: petal width and petal length
 - What does this tell us?



Visualization Techniques: Box Plots

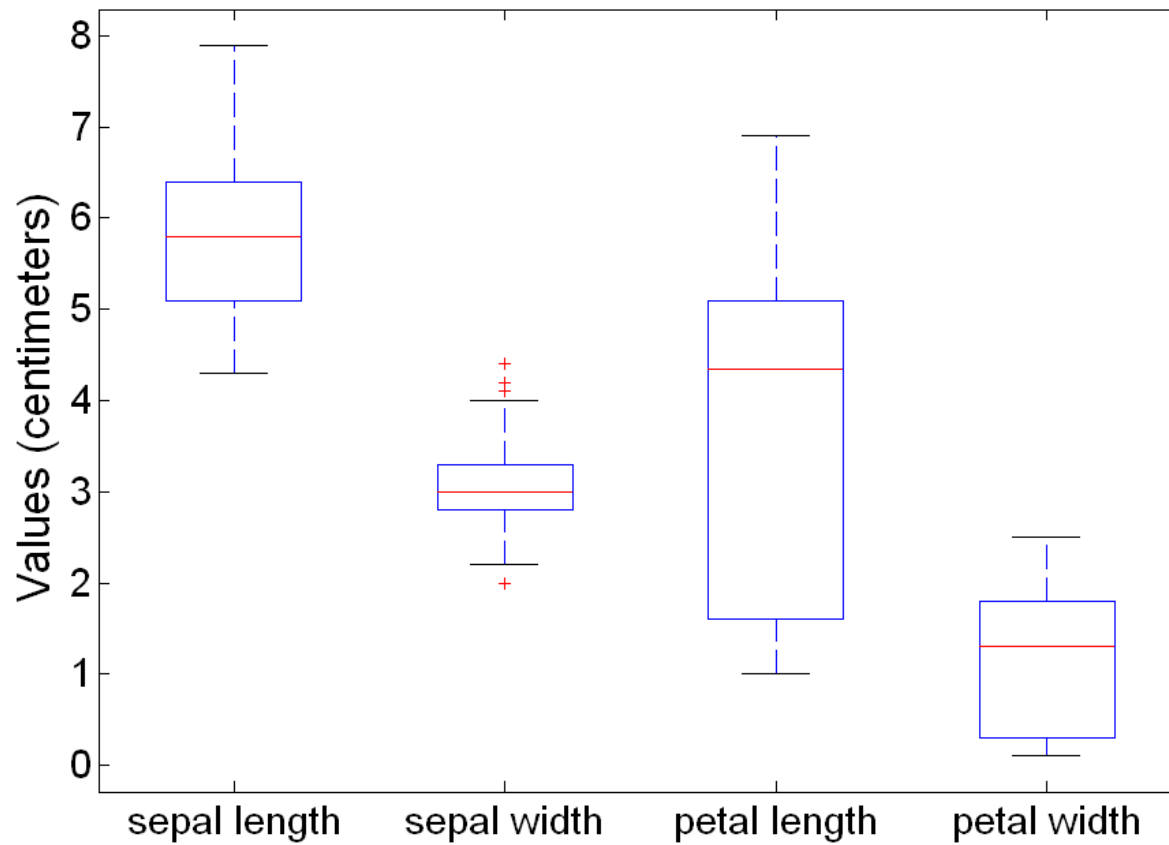
■ Box Plots

- Invented by J. Tukey
- Another way of displaying the distribution of data
- Following figure shows the basic part of a box plot



Example of Box Plots

- Box plots can be used to compare attributes

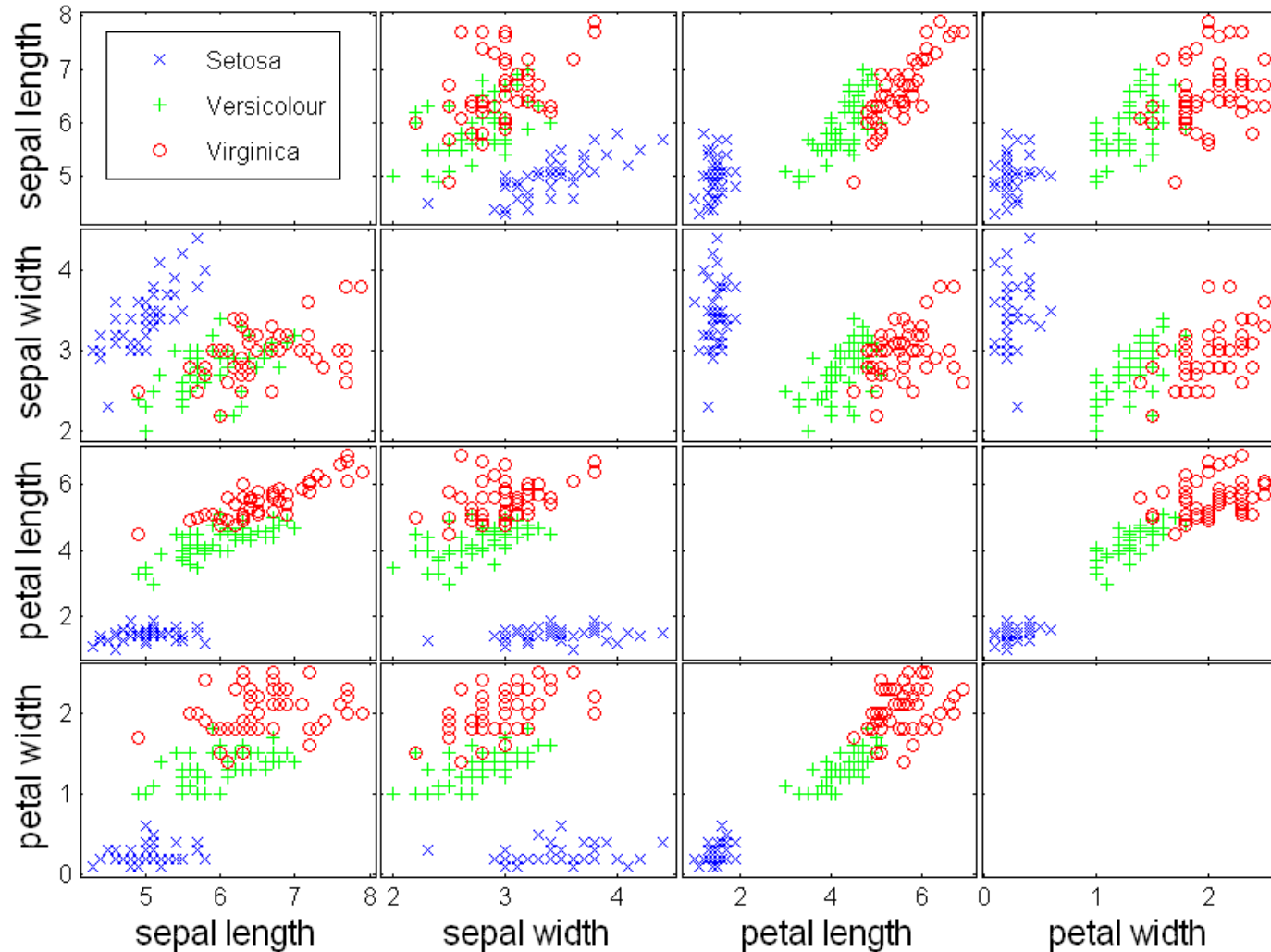


Visualization Techniques: Scatter Plots

■ Scatter plots

- Attributes values determine the position
- Two-dimensional scatter plots most common, but can have three-dimensional scatter plots
- Often additional attributes can be displayed by using the size, shape, and color of the markers that represent the objects
- It is useful to have arrays of scatter plots can compactly summarize the relationships of several pairs of attributes
 - See example on the next slide

Scatter Plot Array of Iris Attributes

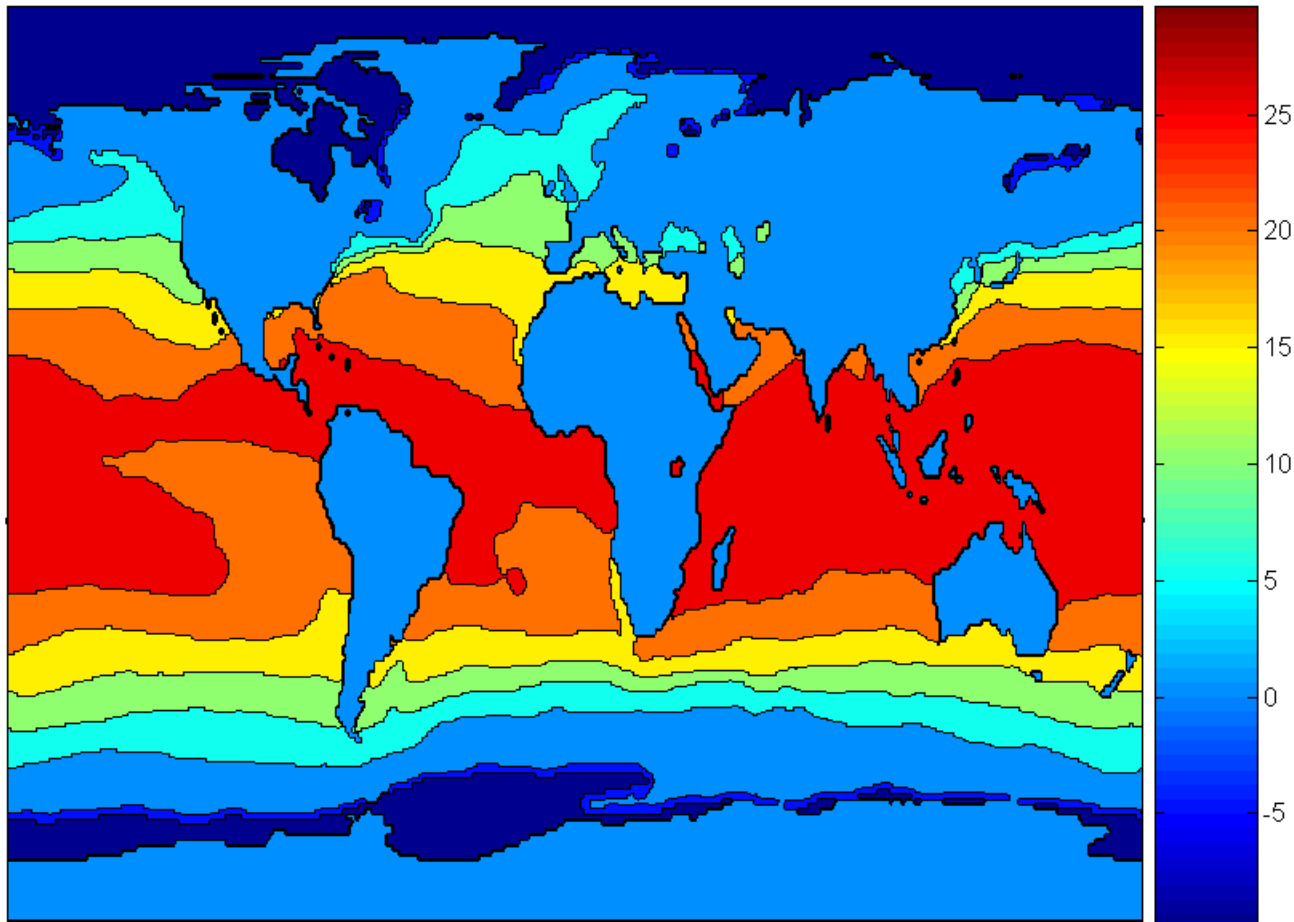


Visualization Techniques: Contour Plots

■ Contour plots

- Useful when a continuous attribute is measured on a spatial grid
- They partition the plane into regions of similar values
- The contour lines that form the boundaries of these regions connect points with equal values
- The most common example is contour maps of elevation
- Can also display temperature, rainfall, air pressure, etc.
 - An example for Sea Surface Temperature (SST) is provided on the next slide

Contour Plot Example: SST Dec, 1998



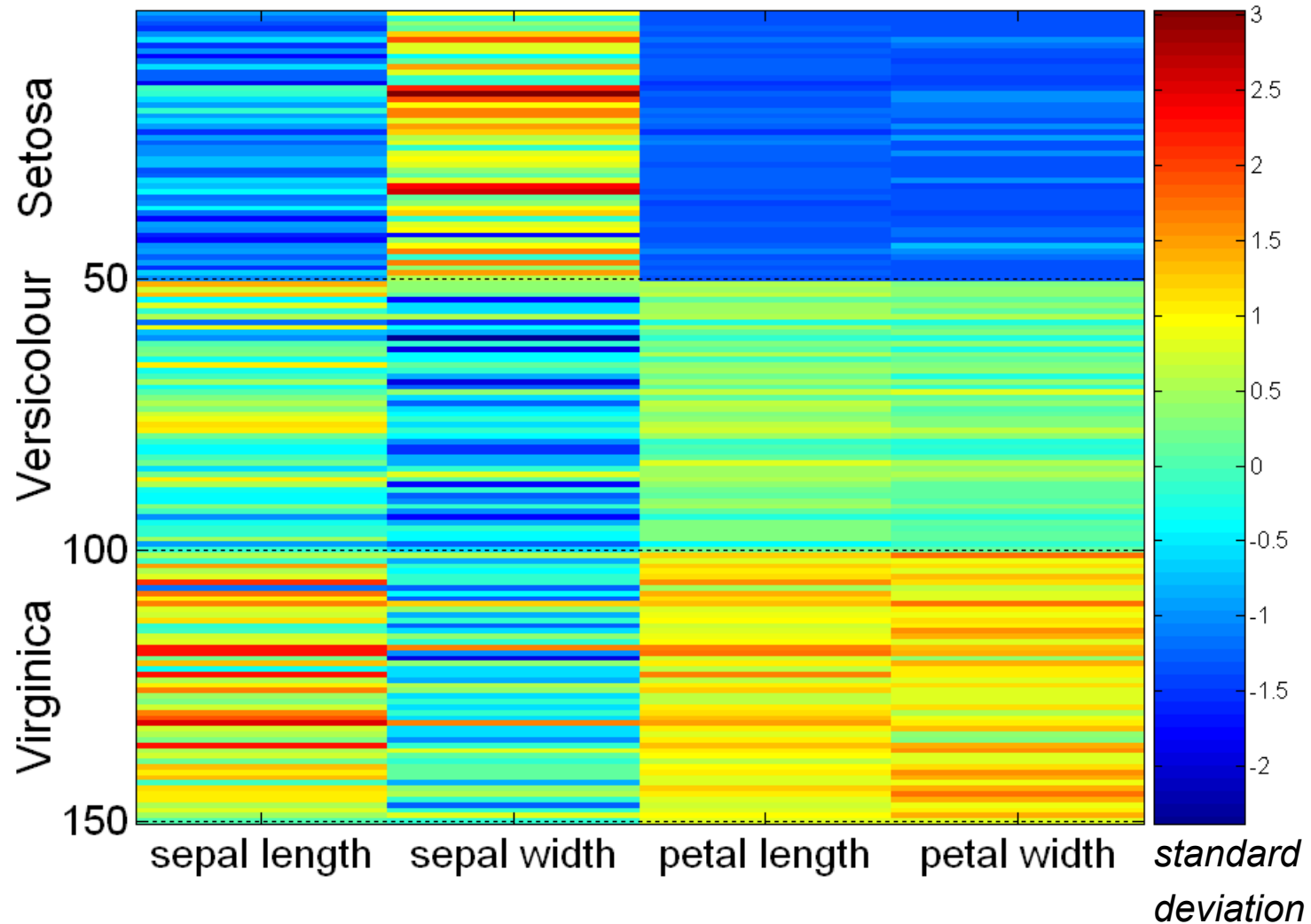
Celsius

Visualization Techniques: Matrix Plots

- Matrix plots

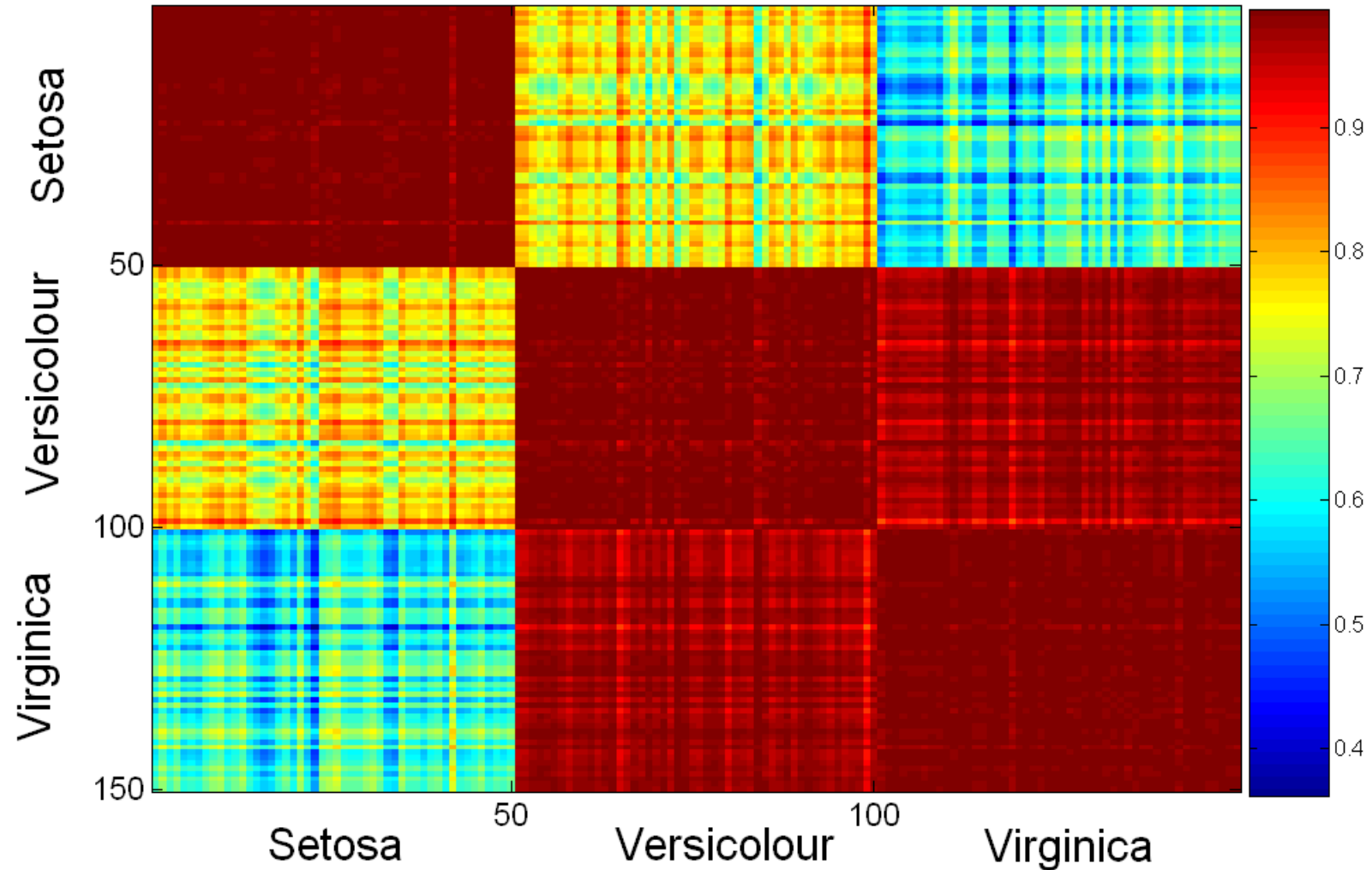
- Can plot the data matrix
- This can be useful when objects are sorted according to class
- Typically, the attributes are normalized to prevent one attribute from dominating the plot
- Plots of similarity or distance matrices can also be useful for visualizing the relationships between objects
- Examples of matrix plots are presented on the next two slides

Visualization of the Iris Data Matrix



columns have been standardized to have a mean of 0 and standard deviation of 1.

Visualization of the Iris Correlation Matrix

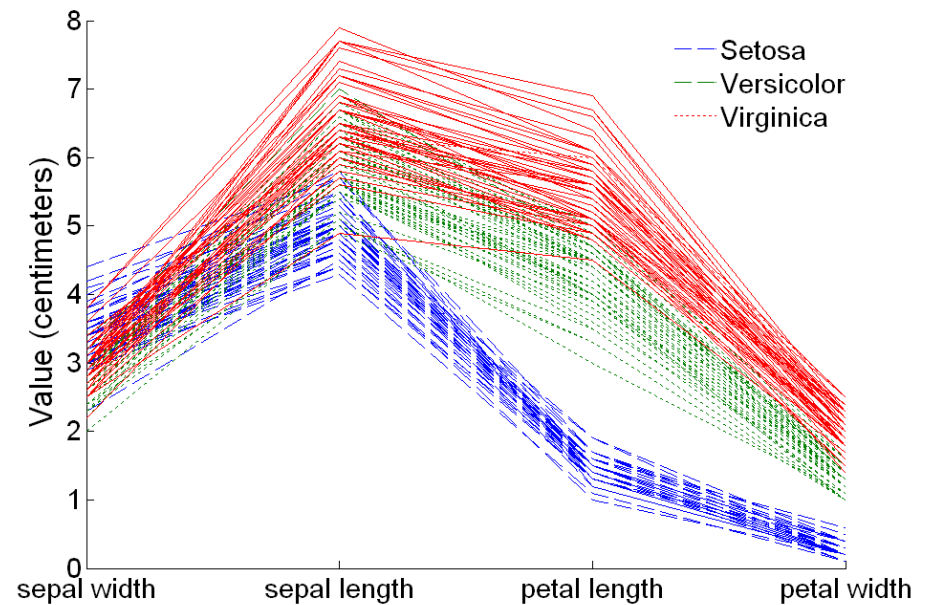
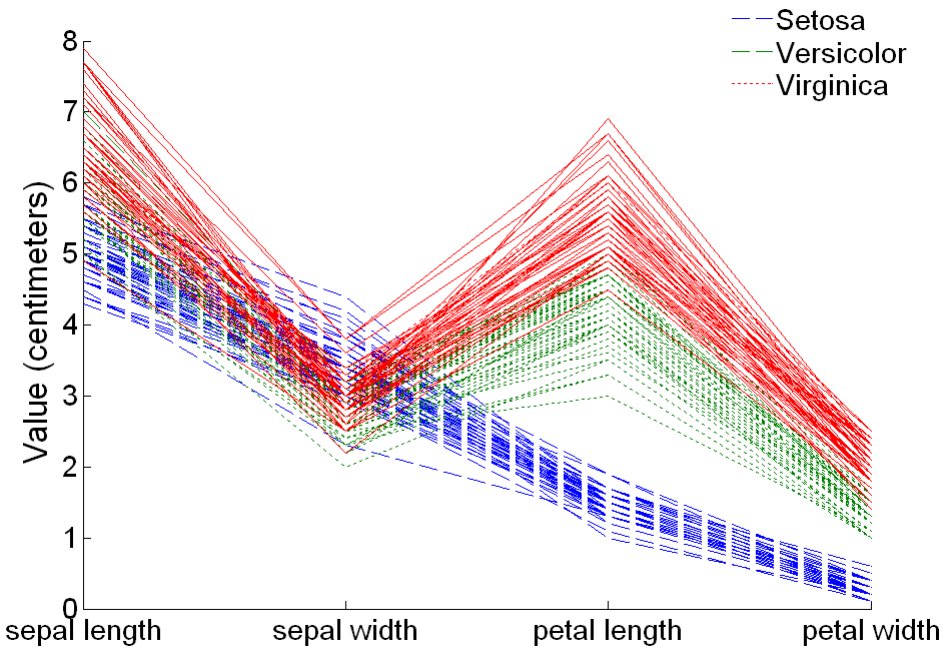


Visualization Techniques: Parallel Coordinates

■ Parallel Coordinates

- Used to plot the attribute values of **high-dimensional data**
- Instead of using perpendicular axes (e.g., scatter plot), use a set of **parallel axes**
- The attribute values of each object are plotted as a point on each corresponding coordinate axis and the **points are connected** by a line
- Thus, each object is represented as a line
- Often, the lines representing a distinct class of objects **group** together, at least for some attributes
- **Ordering** of attributes is important in seeing such groupings

Parallel Coordinates Plots for Iris Data



Other Visualization Techniques

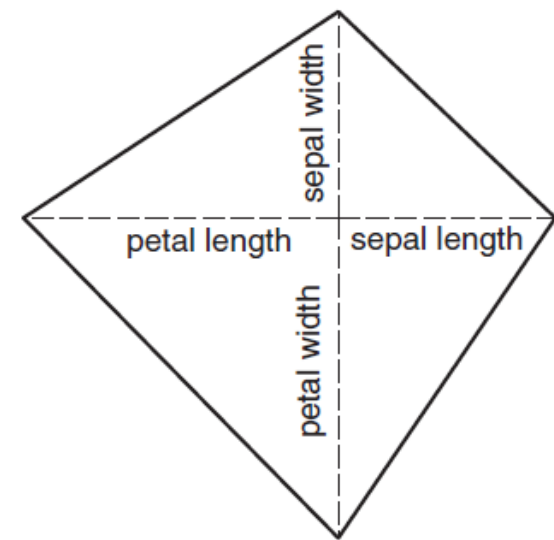
■ Star Plots

- Similar approach to parallel coordinates, but axes **radiate** from a central point
- The line connecting the values of an object is a **polygon**

■ Chernoff Faces

- Approach created by Herman Chernoff
- This approach associates each attribute with a characteristic of a face
- The values of each attribute determine the **appearance** of the corresponding facial characteristic
- Each object becomes a separate face
- Relies on **human's** ability to distinguish faces

Star Plots for Iris Data



1



2



3



4



5

Setosa



51



52



53

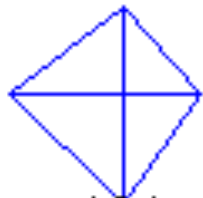


54



55

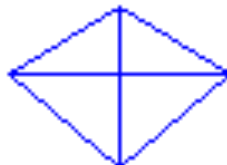
Versicolour



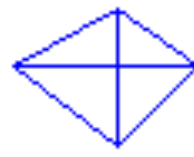
101



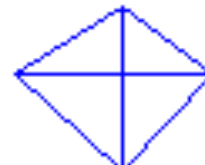
102



103



104



105

Virginica

Chernoff Faces for Iris Data

Data Feature	Facial Feature
sepal length	size of face
sepal width	forehead/jaw relative arc length
petal length	shape of forehead
petal width	shape of jaw



1



2



3



4



5

Setosa



51



52



53



54



55

Versicolour



101



102



103



104



105

Virginica